

Three Bounds on a Null: Testing the Link Between Fast-Weight Write Geometry and In-Context Composition

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Abstract

Fixed-state sequence models bind in-context associations in a matrix-valued fast-weight state; geometric observables have been proposed as lenses on composition. We test whether this observable predicts, or causally improves, in-context multi-hop composition in DeltaNet-family language models spanning 14M to 1.31B parameters, and find a null bounded three ways. First, a state-space composition readout, deployed three structurally different ways, returned a recovered fraction of exactly zero at 366 of 366 readings, gates failing at every cell. A pre-submission positive control caught a state-layout defect in the readout; after the fix, the 320-reading marker-template sub-grid no longer reads zero, but two pre-registered correspondence nulls reproduce the signal at every one of those readings: recovery there is indistinguishable from a broken correspondence; the other two deployments keep their pre-fix zero unconfirmed. Second, a vocabulary-space behavioral contrast bounds any causal effect of a write-geometry intervention below its pre-registered three-seed detection floor of 1.5 to 1.7 loss units; the sole reading meeting the pre-registered differential condition (one of five that exclude zero across forty tests) is a transient, harm-direction deviation. Third, a twelve-seed replication of the transient was stopped by its pre-registered batch-effect gate (cohort variance ratio 4.47 against a heuristic cutoff of 4.0), and the new cohort’s interval spans zero. Each bound names the observation that would overturn it. The result is a null with visible assumptions and closed escape routes.

1 Introduction

Linear-attention and delta-rule language models compress an entire context into a fixed-size matrix state (Schlag et al., 2021; Yang et al., 2024). A companion research program measured geometric observables of the written state (e.g. span fraction, effective key rank) across scales and under targeted interventions, carrying an unexamined promise that write geometry matters for in-context multi-hop composition. We test that promise and find a null. A reader of a null has two standing worries, that the instrument never worked and that the analysis could declare a null after seeing the data. Three pre-registered bounds address both:

1. An instrument bound. A state-space composition readout, deployed three structurally different ways, returned a recovered fraction of exactly zero at 366 of 366 readings under the pre-fix instrument, its validity gates failing at every cell. A pre-submission positive control on the readout’s production path failed outright, exposing a state-layout defect; after the fix, the marker-template sub-grid (320 of the 366 readings) no longer reads zero, but two pre-registered correspondence nulls reproduce the apparent signal at every one of those readings; the other two deployments were not independently re-verified (Section 3). One reading clearing a null, or one premise gate passing, would have voided this bound; none did.
2. A power bound. A vocabulary-space behavioral contrast, with a three-seed detection floor of 1.5 to 1.7 loss units registered before unblinding, leaves three of the four per-arm contrasts unresolved; the sole reading meeting the pre-registered differential condition is

44 a transient, harm-direction deviation. A determinate effect above the floor would have
45 voided this bound.

46 3. A replication bound. A targeted twelve-seed extension of that transient was stopped by
47 its own pre-registered batch-effect gate (a 4.47-fold cohort variance mismatch against a
48 heuristic 4.0 cutoff), and the new cohort’s interval spans zero. A clean pooled interval
49 excluding zero would have voided this bound.

50 The contribution is the bounded null and the measurement discipline that kept it inter-
51 pretable.

52 2 Models, Task, and Instruments

53 Checkpoints and task. All experiments probe DeltaNet-family language models (Yang et al.,
54 2024) pretrained on a reasoning-dense corpus mix (OpenR1-Math) and an encyclopedic mix
55 (WikiText-103, Merity et al., 2017). An intervention grid (14M parameters) varies a frozen-
56 key-bias intervention on the fast-weight write path in three validation-loss-matched arms:
57 control, per-token, and global. A scale ladder spans 14M, 98M, 392M, and 1.31B parameters
58 (state dimension 64, 64, 128, 128; depth 2, 12, 16, 22). Each probe episode binds K entities
59 into one directed cycle with bind statements, then queries the entity $h \in \{1, \dots, 4\}$ hops
60 around the cycle; a full K -cycle prevents held-out hop depths from collapsing modulo short
61 cycle lengths. Load is K relative to state dimension ($K \in \{20, 32\}$ on the grid; near-capacity
62 per rung).

63 Instrument 1: a geometric readout. For a layer’s terminal state S_T and effective query q_{eff} ,
64 the readout predicts the h -hop target as $S_T^h q_{\text{eff}}$ and scores exact continuous recovery: the
65 fraction of queries reaching cosine at least 0.9 with the target value vector. Continuous
66 recovery, rather than argmax over candidates, follows Nichani et al. (2024), under which
67 argmax decoding can succeed without the state carrying the claimed structure. Registered
68 validity gates accompany the readout: an $h=1$ sanity floor and two alignment premises, (iii)
69 query–key and (iv) key–value, each requiring a same-entity median cosine above a shuffled-
70 entity null’s 95th percentile (Appendix A); gate failure routes to probe-invalid, never to
71 refutation. A correspondence-null battery (Section 3) screens any nonzero reading against
72 a deliberately wrong target.

73 Instrument 2: a behavioral contrast. A familiarization phase trains checkpoints for 5,000
74 steps on the bind/query task mixed into the pretraining corpus; the contrast is the held-out-
75 episode query loss L_q (a vocabulary-space cross-entropy through the language-model head,
76 machinery disjoint from Instrument 1), differenced control minus arm at five checkpoints.
77 A six-outcome trajectory classification and a differential condition (the $K=32$ effect deter-
78 minate while $K=20$ is not) were fixed before unblinding. Every wave ran gate-first, with
79 verdicts routed mechanically (Appendix A).

80 3 The Geometric Readout Never Fires

81 The readout was defective, and the fix did not save it. A pre-submission positive control
82 on the readout’s production state-extraction function failed outright (0 of 256 recovered
83 at every hop), exposing a $[K, V]$ -vs- $[V, K]$ state-layout transpose; after a one-line fix and
84 an independent review, the same control recovers 1.0000 at every hop (Appendix A). Re-
85 running the wave-1/wave-2 grid (80 cells, 320 readings) through the corrected readout no
86 longer returns zero: 78 readings are nonzero, one $h=1$ cell reaching 0.87. Two pre-registered
87 correspondence nulls – a label-shuffle null, later found vacuous at $h=1$ by construction, and
88 a positional derangement null with teeth at every hop – both reproduce this apparent signal
89 at every one of the 320 readings (0/320 clear either bar), tracing it to collapsed prediction
90 directions scored against a near-collinear value population (Appendix A). Bound 1’s claim
91 is that recovery is statistically indistinguishable from a broken correspondence, not that it
92 reads zero; the alignment-premise gates (Figure 1) fail at all 320 readings regardless, before
93 and after the fix.

Wave (instrument)	Grid	Readings	Registered outcome
1 zero-shot, marker [‡]	78 cells, 14M–392M	312/312 = 0.0	probe-invalid
2 zero-shot, 1.31B rung [‡]	2 cells	8/8 = 0.0	probe-invalid
1+2 fixed-lens re-metric	same 80 cells	78/320 nonzero, $h=1 \leq 0.87$	unadjudicated
1+2 correspondence nulls	same 80 cells	0/320 clear either of 2 nulls	null-indistinguishable
3 zero-shot, natural lang. [§]	4 gate cells	16/16 = 0.0	probe-invalid, refused
4 familiarized gate [§]	6 cells $\times$ 5 ckpts	30/30 = 0.0; L_q falls	dissociation
5 behavioral contrast	18 cells, 4 contrasts [†]	3 unresolved, 1 transient	power-bounded
6 seed extension, $n=12$	18 pretrain cells	variance ratio 4.47	batch-effect-flagged

Table 1: The program at a glance. Waves 1–4 deploy the geometric readout (recovered fraction at cosine ≥ 0.9). Waves 5–6 deploy the vocabulary-space contrast; the one determinate transient is in the harm direction and its $n=12$ replication was stopped by the pre-registered batch-effect gate. [†]Per-arm re-derivation; corpus-level returned both unresolved. [‡]Raw reading superseded below; independent premise failure still routes to probe-invalid. [§]Identical pre-fix function, not independently re-run (Appendix A).

94 Zero-shot, natural language (wave 3). Two template families – a gift-verb family with the
95 query marker dropped, and a corpus-native succession-verb family – fail identically: 16 of
96 16 readings at 0.0, the $h=1$ gate false at all 4 cells, not a surface-form artifact.

97 Task-familiarized (wave 4). The strongest-case test trains control checkpoints on the task
98 itself for 5,000 steps and applies the gate at five checkpoints per cell. It fails at 30 of 30
99 readings, 0 of 512 queries recovered each time, while the vocabulary-space query loss falls
100 21.8 to 46.4 percent (mean 35.9 percent) in 6 of 6 cells, from checkpoint 250 (first post-
101 warmup) to checkpoint 5,000 (Figure 2, Appendix B): the models measurably learn the task
102 through one readout while the premise gates fail throughout and the recovered-fraction floor
103 – a pre-fix reading, Appendix A – stays at 0.0. The dissociation, on the premise gates rather
104 than the pre-fix floor, indicts the readout construct, not the models (partial task learning,
105 short of the registered 50-percent pin). Waves 3–4 share the pre-fix function and were not
106 independently re-run; their probe-invalid routing rests on the unaffected premise gates alone,
107 not the raw zero reading.

108 Gates with enforcing code paths. Wave 1’s chain computed its abort gate but never acted
109 on it, running the full grid past an already-invalid probe; every later chain enforced its gate
110 at the process boundary, and waves 3 and 4 each refused launch on a real failing gate.

111 4 The Behavioral Contrast Is Power-Bounded

112 Wave 5 promotes the vocabulary-space contrast to the primary instrument: all three arms
113 are familiarized under identical recipes (18 cells), and the arm effect is the held-out query-
114 loss difference, control minus arm, so positive means the intervention helps.

115 The registered power floor. Before unblinding, the three-seed detection floor was derived
116 from the control cells’ between-seed spread: a σ proxy of 0.43 to 0.48 loss units gives a
117 95-percent half-width of roughly 1.5 to 1.7. For calibration, the entire familiarization effect
118 is about 1.69 loss units: the instrument detects only effects about as large as everything the
119 model learns in the phase, so the registration anticipated unresolved outcomes, to be read
120 as measurement bounds, never as evidence of no effect.

121 Results. The registered corpus-level classifier, which uses the global arm as each corpus’s
122 representative, returns both corpora unresolved; a disclosed build-time per-arm re-derivation
123 (folded into the analysis code, re-validated against the archived deltas; Appendix A) splits
124 this into four contrasts, three unresolved. The fourth (encyclopedic corpus, per-token arm)
125 classifies as transient: at checkpoint 2,500 the $K=32$ effect is determinate at -0.500 , interval
126 $[-0.624, -0.376]$, while $K=20$ is not, and by checkpoint 5,000 it dissolves (Figure 3, left,
127 Appendix B). A held-out-hop secondary readout fires at the same cell and checkpoint, same
128 direction, consistent with one coherent mid-training deviation.

129 Sign discipline and multiplicity. Of 40 uncorrected 95-percent intervals in the primary table,
 130 five exclude zero against two expected by chance, all in the harm direction (arm loss above
 131 control); four of the five sit at checkpoint 1,000, both loads of two different cells, consistent
 132 with checkpoint-level sampling noise (Appendix A). Only the fifth, at checkpoint 2,500, also
 133 satisfies the pre-registered differential condition and draws the held-out-hop corroboration;
 134 that compound condition, not any single interval, is the one determinate signal. “The
 135 intervention matters” ignores that transients were registered as training-dynamics artifacts;
 136 “no effect” ignores the power floor.

137 5 The Replication Gate Refuses the Pool

138 Wave 6 extends the transient’s anchor cell (encyclopedic corpus, per-token arm, $K=32$,
 139 checkpoint 2,500) to twelve seeds with three registered integrity mechanisms: nine fresh
 140 pretraining seeds per arm (18 cells), an archived-values loader whose runtime guard forbids
 141 live re-scoring, and a pre-pooling batch-effect gate (means within twice the pooled standard
 142 error; variance ratio below 4.0).

143 The gate fired. The archived cohort’s between-seed standard deviation (1.154) is 2.1
 144 times the new cohort’s (0.546): variance ratio 4.47, past the heuristic 4.0 cutoff (a rout-
 145 ing threshold, not a calibrated test; the one-sided 5-percent $F(2,8)$ value is ≈ 4.46); the
 146 means agree (shift 0.364, threshold 1.382). The registered routing reports the cohorts sep-
 147 arately, batch-effect-flagged, with no decision-grade pooled interval: the archived cohort
 148 reproduces its original $[-0.624, -0.376]$ by construction; the new nine-seed cohort’s own in-
 149 terval, $[-0.506, +0.357]$, spans zero; and the naive twelve-seed pool, diagnostic only, reads
 150 $[-0.509, +0.147]$, a refutation absent the gate (Figure 3, right, Appendix B).

151 What stands. The transient is neither confirmed nor refuted; it keeps its three-seed weight.
 152 Every computable pooled interval, at 7 of 10 primary and 7 of 10 held-out (checkpoint, load)
 153 pairs, contains zero. The variance mismatch is not established as systematic; its direction
 154 flips across checkpoints, consistent with small- n imprecision in the archived cohort.

155 6 Related Work

156 Zoology (Arora et al., 2023) and Based (Arora et al., 2024) establish the recall-state-size
 157 tradeoff and the ladder pattern this program follows; both train architectures for the probe
 158 setting, where this program probes general-purpose checkpoints post hoc and reports in-
 159 strument validity, not a capacity frontier. Okpeke & Orvieto (2025) show that optimizer
 160 choice, not architecture alone, drives much of the reported Transformer-recurrent recall
 161 gap; this paper bounds a different axis, write-geometry validity, and makes no positive
 162 mechanism claim. Grazi et al. (2024) prove positive-eigenvalue DeltaNet cannot compose
 163 certain groups (e.g. parity), a ceiling lifted only by negative-eigenvalue transitions; Merrill
 164 et al. (2024) give the analogous TC^0 ceiling for diagonal SSMs. Nichani et al. (2024) supply
 165 the associative-memory analysis behind the exact-recovery readout (Section 2). Reporting
 166 follows the pre-registration workshop (Bertinetto et al., 2021) and negative-results tradition
 167 (Forde et al., 2020).

168 7 Scope of the Bounds

169 Bounded. The single-layer self-iteration readout carries no signal distinguishable from a
 170 broken correspondence at every scale and corpus tested against two correspondence nulls
 171 (Section 3); intervention effects beyond 1.5 to 1.7 loss units are excluded, and the transient
 172 is bounded in duration, direction, and weight.

173 Not bounded. The hypothesis survives below the power floor and through cross-layer hand-
 174 off; an expressivity ceiling (Grazi et al., 2024) may also contribute, though wave 4 points
 175 toward the observable. Nothing transfers beyond the DeltaNet family or bind/query gram-
 176 mar. A gate without an enforcing code path is only a sentence; unblinded routing kept zeros
 177 and transients from becoming headlines.

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A Registered Gates and Routing Rules

This appendix reproduces, in compact form, the pre-registered gates and mechanical routing rules the main text relies on. Each was fixed in the program’s design registry before the corresponding data existed.

Geometric-readout validity gates (waves 1–4). A cell’s readout is valid only if all of the following hold. (a) $h=1$ sanity floor: the one-hop recovered fraction must exceed the upper edge of a shuffled-pairing null by more than the null’s width, and must also exceed an absolute floor of 0.10. (b) Premise (iii): the median cosine between an entity’s query projection and the same entity’s key projection must exceed the 95th percentile of a shuffled-entity null. (c) Premise (iv): the median cosine between the key and value projections of the same token must exceed the same null percentile. Routing: failure of any gate routes the cell to probe-invalid; the registered rule states that gate failure never routes to refutation. The readout’s recovery math (state-power application and the cosine-recovery scorer) is unit-tested against hand-built matrices.

The positive control (2026-07-11): failed, root-caused, fixed, independently reviewed. A pre-submission positive control was run end-to-end through a real forward pass on the production multi-layer kernel path, closing the gap the previous paragraph flagged as open. A synthetic $K=32$ mutually-orthonormal-key single cycle was injected at the readout layer of

a real trained checkpoint (mutually orthonormal keys make the correct h -hop answer exactly the $(a+h) \bmod K$ key, so recovery is analytically known), scored through the identical production squeeze_state_head $\rightarrow$ apply_state_power $\rightarrow$ cosine-recovery code path every grid cell uses. Result. The production path recovered 0 of 256 at every h (cosine indistinguishable from noise), while a manually-transposed variant of the identical state recovered 1.0000 at every h through the same scoring code; diagnostics ruled out a script artifact (key orthonormality to 3.6×10^{-7} , zero hook-injection deviation, no near-zero β items, and the real kernel's raw state matches an independently implemented reference recurrence to 0.17% relative Frobenius error). Root cause. A single-write closed-form adjudication (one basis-vector key/value pair, all other positions state-neutral, computed independently of any probe code) confirmed squeeze_state_head returned the underlying kernel's raw $[K, V]$ (key-major) state layout unchanged, while apply_state_power contracts its query vector against what it assumes is the last (key) axis of a $[V, K]$ (value-major) layout: the readout output matched the $[K, V]$ closed form at 0.0 relative Frobenius error and the $[V, K]$ closed form at exactly $\sqrt{2}$ (the pure-transpose signature), and the two closed forms were themselves exact transposes of each other. This is the same defect class as a previously corrected defect elsewhere in this program's codebase, never inherited here because this readout implements an independently written state-extraction function. Fix and review. The fix is a one-line transpose, final_state[:, 0, :].transpose(-1,-2) in place of the untransposed slice, mirroring the already-corrected reference function's own transpose exactly. A hand-computed regression test asserts both the post-fix recovery and, as a kill-proof, that the pre-fix layout inlined still fails the identical assertion. An independent review (a fresh reviewer with no access to the fixing session's own reasoning) confirmed the diff is minimal and correctly scoped: squeeze_state_head has exactly one production call site feeding apply_state_power (the intended target) and the state condition-number covariate, which is provably transpose-invariant ($\text{svdvals}(S) = \text{svdvals}(S^T)$), so no other production code path changes behavior. Re-running the positive control under the fixed function recovers 1.0000 at every h (cosine ≈ 0.99999), and the manually-transposed arm, now playing the pre-fix-equivalent wrong-layout role, recovers 0.0000: the exact role-swap the closed-form adjudication predicts.

The re-metric (78/320 nonzero) and why it does not reopen the lane. Re-running the full wave-1/wave-2 grid – 80 cells, 320 (cell, h) readings, identical archived checkpoints and deterministic episode seeds, one forward pass per cell – under the fixed function does not reproduce the pre-fix zero: 78 readings are nonzero (30 of 60 intervention-grid cells, 8 of 20 scale-ladder cells – the re-metric's own Leg-A/Leg-B partition, a different axis from Table 1's wave-1 (78 cells) / wave-2 (2 cells) split of the same 80 cells – carry ≥ 1 nonzero hop), $h=1$ recovered fraction reaching 0.8691 at its strongest cell (a per-token-arm wikitext cell, seed 1, $K=32$) and $h \geq 2$ reaching 0.6375 in one cell. The alignment-premise gates read key/query/value hook values, never the state-extraction output the defect touched, and continue to fail at all 320 readings (0/320 both premises) before and after the fix, so under the registered routing rule this reading alone licenses no confirm regardless of what follows.

Two correspondence nulls, pre-registered before either ran, were used to test whether the fixed-lens signal reflects genuine state-target correspondence or is trivially satisfiable by any (state, wrong-target) pairing. The first, a label-shuffle null (the readout's existing null-relative machinery, extended from one previously-exercised calibration cell to all 80 grid cells), grades the identical forward pass's prediction against an independently drawn Hamiltonian K -cycle rather than the true one. This null reproduces the real reading at every one of the 320 readings (0/320 clear the pre-registered bar: null $<$ half the real value, AND real exceeds the null's bootstrap upper edge by the null's own width, AND real ≥ 0.10); at $h=1$ specifically, real and null read identically at every strong cell (e.g. 0.8691/0.8691), traced to a construction property rather than a grading bug: the $h=1$ target is computed with zero permutation iterations, so it never consults the shuffled cycle at all, making the label-shuffle null vacuous at $h=1$ by construction in every cell, including the readout's own original calibration cell (a healthy-draw check separately confirmed the null cycle itself agrees with the true one on only 4.1% of entries, ruling out a degenerate shuffle). A second, positional derangement null was therefore pre-registered as the anticipated fallback: it grades the same prediction against a fixed-point-free random permutation of the candidate value

slots, breaking correspondence at every hop including $h=1$. This null also reproduces the fixed-lens signal at every one of the 320 readings (0/320 clear; the strongest cell, 0.8691 real, reads 0.8125 against the deranged target, 94% reproduction against the pre-registered 50% bar), and was itself kill-proofed with a perfect-correspondence fixture (real 1.0, deranged 0.0) and a collapsed-geometry fixture (real $\approx$ deranged) before being trusted.

The mechanism both nulls point to is geometric, not compositional: at the readout's strongest cells the cross-checkpoint prediction direction converges (mean pairwise cosine 0.9996 across queries, under state condition numbers of 1.8×10^4 to 2.1×10^6) while the candidate value population is itself near-collinear (mean pairwise cosine 0.9648); a near-constant prediction direction scores cosine above 0.9 against nearly any member of a near-collinear population, real or deranged target alike. Provenance flag: these two covariates (0.9996, 0.9648), and the 4.1-percent healthy-draw check above, are a design-registry record of an on-box diagnostic on the single strongest cell, not backed by a separately archived raw artifact. No raw state tensors were saved by the original harvest to recompute them from, so the empirical verdict above (0/320 clear either null) does not depend on them, but this specific mechanistic interpretation of why does, and is reported with that provenance disclosed rather than independently re-derived. An independent episode-resampling check (a fresh deterministic episode draw on the identical checkpoints) confirms the underlying bimodality – which cells carry any signal at all – is a stable property of the trained state (20 of 21 floor-clearing cells stay on under a resampled episode), not an artifact of one lucky draw; within this family, the deployed per-token intervention arm concentrates the effect (20 of 72 $h \geq 2$ readings clear a 0.10 floor, versus 1 of 36 for the off arm and 0 of 72 for the global arm): concentration of the artifact, not of composition, per the same two nulls.

The corrected claim, binding throughout this paper. Geometric recovery is statistically indistinguishable from a broken correspondence at every scale, corpus, and hop the corrected instrument was run on. It is not that recovery reads exactly zero, which was true only of the pre-fix instrument on this sub-grid, and it is not a confirmed correspondence, which both correspondence nulls exclude. Waves 3 and 4 (natural-language and task-familiarized deployments, Section 3) call the identical pre-fix state-extraction function and were not independently re-run under the fix or tested against either correspondence null; their probe-invalid routing rests entirely on the alignment-premise gates, which the defect never touched and are therefore unaffected, but their own raw zero readings carry the same unconfirmed status the wave-1/wave-2 zero readings carried before this re-verification. That status is reported here as disclosed, not assumed corrected by extension.

Behavioral-contrast classification (wave 5). The per-checkpoint arm effect is $\Delta(c) = L_q(\text{control}, c) - L_q(\text{arm}, c)$, with three-seed 95-percent intervals ($t(2, .975) = 4.303$). A checkpoint satisfies the differential condition when the $K=32$ interval excludes zero, the $K=20$ interval does not, and $|\Delta_{32}| > |\Delta_{20}|$. Trajectories classify into six registered outcomes: persistent, transient (the condition holds at an interior checkpoint and fails at the terminal one), late-emergent, converged-equivalent, unresolved, and a familiarization-null exclusion. The three-seed power floor (a detectable effect of roughly 1.5 to 1.7 loss units, derived from the control cells' between-seed spread) was registered together with the instruction that unresolved outcomes be reported as measurement bounds.

The registered pipeline computed one classification per corpus, using the global arm as that corpus's representative signal, and classified both corpora unresolved. The four-way (corpus, arm) breakdown reported in the main text and Table 1 applies the same registered classification primitives to each arm's own trajectory. This finer split was a build-time re-derivation performed after the corpus-level output was seen; it was later folded into the analysis code and re-validated against the archived per-cell deltas, reproducing the same four verdicts. It is disclosed here as a re-derivation, not presented as the registered corpus-level verdict.

The primary table computes 40 uncorrected 95-percent intervals (4 contrasts $\times$ 5 checkpoints $\times$ 2 loads); five exclude zero: both loads of reasoning-dense $\times$ global at checkpoint 1,000 ($K=32$: $\Delta = -0.203$, $[-0.402, -0.004]$; $K=20$: $\Delta = -0.199$, $[-0.325, -0.074]$), both loads of encyclopedic $\times$ per-token at checkpoint 1,000 ($K=32$: $\Delta = -0.168$, $[-0.301, -0.036]$; $K=20$: $\Delta = -0.109$, $[-0.211, -0.006]$), and the $K=32$ interval of encyclopedic $\times$ per-token

338 at checkpoint 2,500 ($\Delta=-0.500$, $[-0.624, -0.376]$, the reported transient). Only the last
339 also satisfies the compound differential condition above and is corroborated by the indepen-
340 dent held-out-hop readout at the same cell and checkpoint; that compound condition, not
341 any single interval exclusion, is what the main text calls the one determinate signal. The
342 checkpoint-1,000 cells fail the differential condition precisely because their $K=20$ intervals
343 also exclude zero.

344 Replication integrity gates (wave 6). Archived seeds load from stored values only; a runtime
345 guard raises on any live re-scoring of an archived seed. Before pooling, a batch-effect gate
346 requires (a) cohort mean shift below twice the pooled standard error (Welch form) and (b)
347 cohort variance ratio below a pre-registered heuristic 4.0 (not a calibrated F -test; the one-
348 sided 5-percent $F(2,8)$ critical value for the archived three-seed versus new nine-seed cohorts
349 is ≈ 4.46). On gate failure the registered routing reports both cohorts separately, produces
350 no decision-grade pooled interval, and classifies the wave as batch-effect-flagged, outside the
351 four-outcome partition (confirmed at magnitude, confirmed smaller, refuted, new pattern)
352 that a passing gate would have entered.

353 B Supplementary Figures

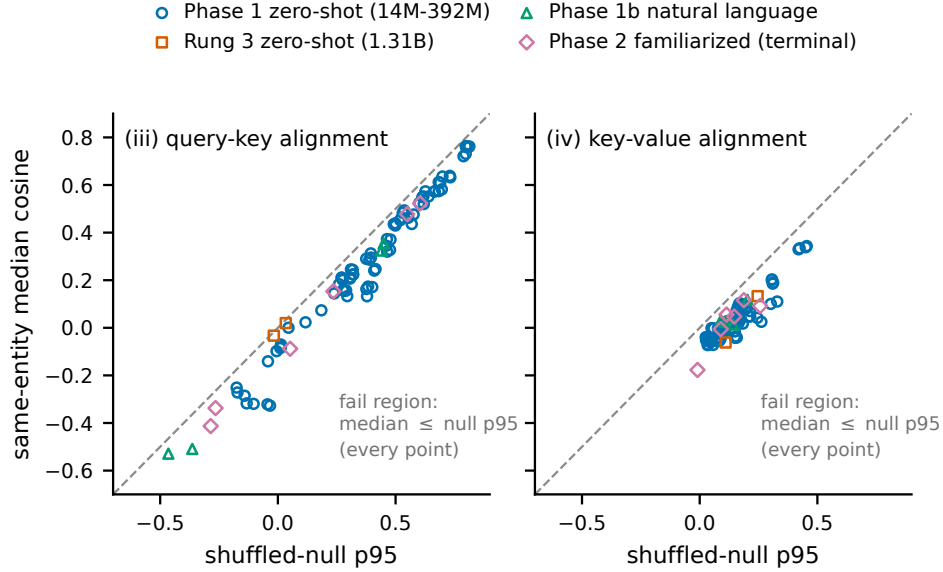


Figure 1: Validity-gate failure at every cell of every instrument. Each point is one cell’s same-entity median cosine for premise (iii) (query–key, left) or premise (iv) (key–value, right) against that cell’s shuffled-entity null p95; passing requires a median above the dashed identity line. All 90 cells across the four geometric-readout waves fall on or below the line, at every scale from 14M to 1.31B and under every surface form and training regime.

354 C Reproducibility

355 Every number in this paper is computed from archived raw result files (per-cell JSON
 356 records written by the experiment chains at run time), not from summaries. All figures
 357 regenerate from one versioned script that loads only those archived files and asserts the
 358 md5 checksum of every file it reads against a recorded manifest, so a changed or substituted
 359 artifact fails the build rather than silently altering a panel. As a compute disclosure for
 360 this venue’s small-scale remit: all experiments ran on one or two GPUs at a time, and the
 361 six waves consumed approximately 9.5 GPU-hours in total. The instrument re-verification
 362 (Section 3, Appendix A) – the positive control, the closed-form adjudication, the 80-cell
 363 re-metric, and both correspondence-null sweeps – consumed a further ≈ 0.52 GPU-hours on
 364 one GPU, disclosed separately since it postdates the original six-wave harvest. The archived
 365 raw files, the figure script with its checksum manifest, the experiment chains, and the full
 366 pre-registration text will be released with the camera-ready version.

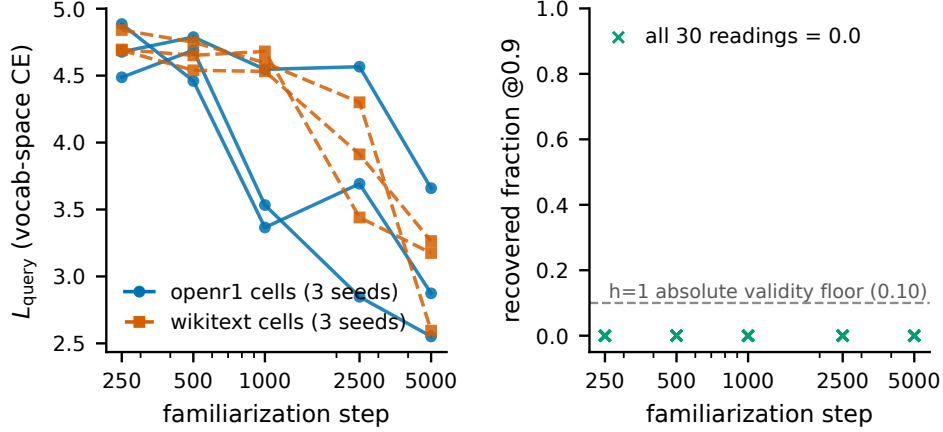


Figure 2: Vocabulary/geometry dissociation under task familiarization (wave 4). Left: held-out in-context query loss L_q for all six control-arm cells (two corpora, three seeds) falls by 21.8 to 46.4 percent between checkpoint 250 (the first post-warmup reading) and checkpoint 5,000. Right: the geometric gate’s recovered fraction at the same five checkpoints of the same cells is exactly 0.0 at all 30 readings, never approaching its 0.10 validity floor (dashed). The model learns the task through one readout while the other stays at floor. The right panel was computed through the pre-fix state-extraction function (Section 3, Appendix A) and has not been independently re-run under the corrected instrument or checked against a correspondence null; the dissociation’s behavioral half (left panel) does not depend on this and is unaffected.

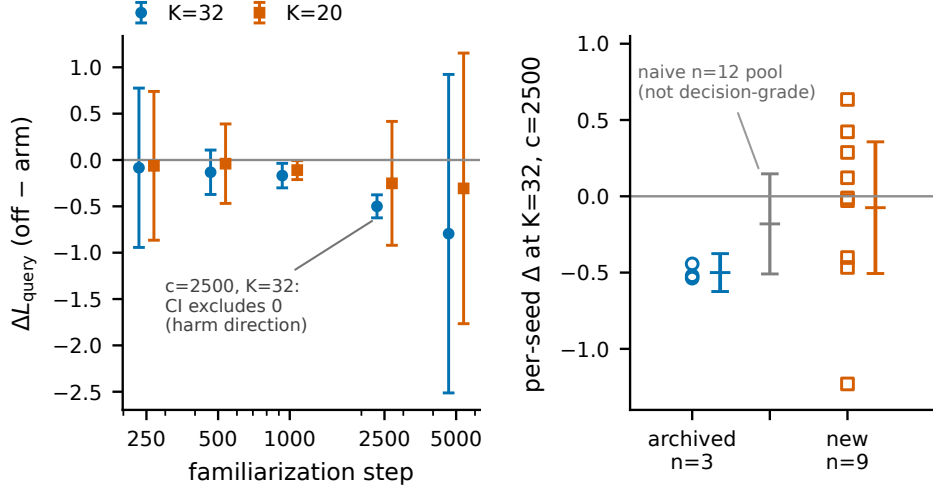


Figure 3: The transient and its stopped replication. Left: per-checkpoint arm effect ΔL_q (control minus per-token arm, encyclopedic corpus) with three-seed 95-percent intervals at both loads; the single reading satisfying the registered differential condition is at checkpoint 2,500, $K=32$ (-0.500 , $[-0.624, -0.376]$), in the harm direction, against a non-determinate $K=20$ reading at the same checkpoint (mean -0.252 , $[-0.920, +0.416]$), and dissolves by checkpoint 5,000 (mean -0.795 , $[-2.513, +0.923]$). Right: the same anchor cell at $n=12$; the archived three-seed cohort (circles, interval $[-0.624, -0.376]$) and the new nine-seed cohort (squares, interval $[-0.506, +0.357]$) fail the pre-registered variance-ratio gate (4.47 against a pre-registered heuristic cutoff of 4.0; a calibrated one-sided 5-percent $F(2,8)$ test would itself require ≈ 4.46), so the naive pooled interval (gray, dotted) is diagnostic only and no decision-grade pooled verdict exists.